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Fall 2025

Homework 8

Written Scenarios

Scenario 1

A player is instantiated in a room and receives information on what is present in the room. The room replies that the east direction is open for navigation, but the north and west directions are impassable while the south is blocked by a monster. The monster does damage of –5 to the player. The player navigates through the east passage to another room. In this room, there are fixtures including a bookcase, a table, and a chair. There are two items present in the room: a piece of paper with writing on it and a feather. There is also a computer that asks the player to enter a password, and its message obscures the room description. The passage that he entered through is now the west passage from the current room. There are no additional open passages to other rooms from this room. Next, the player adds the paper and feather items to his inventory. Finally, the player travels back through the western passage that he arrived from, going back to the starting room.

Back in the original room, the player is given the same information about a monster blocking the north and west passages. The monster attacks the player, causing the player to lose five points of health. Then, the player takes out the paper with magic writing on it and uses it on the monster. The monster is defeated by magic words written on the paper and becomes inactive. The player receives 100 points. The monster now has no effect on the room, and the northern and western passages are now open for the player to travel through.

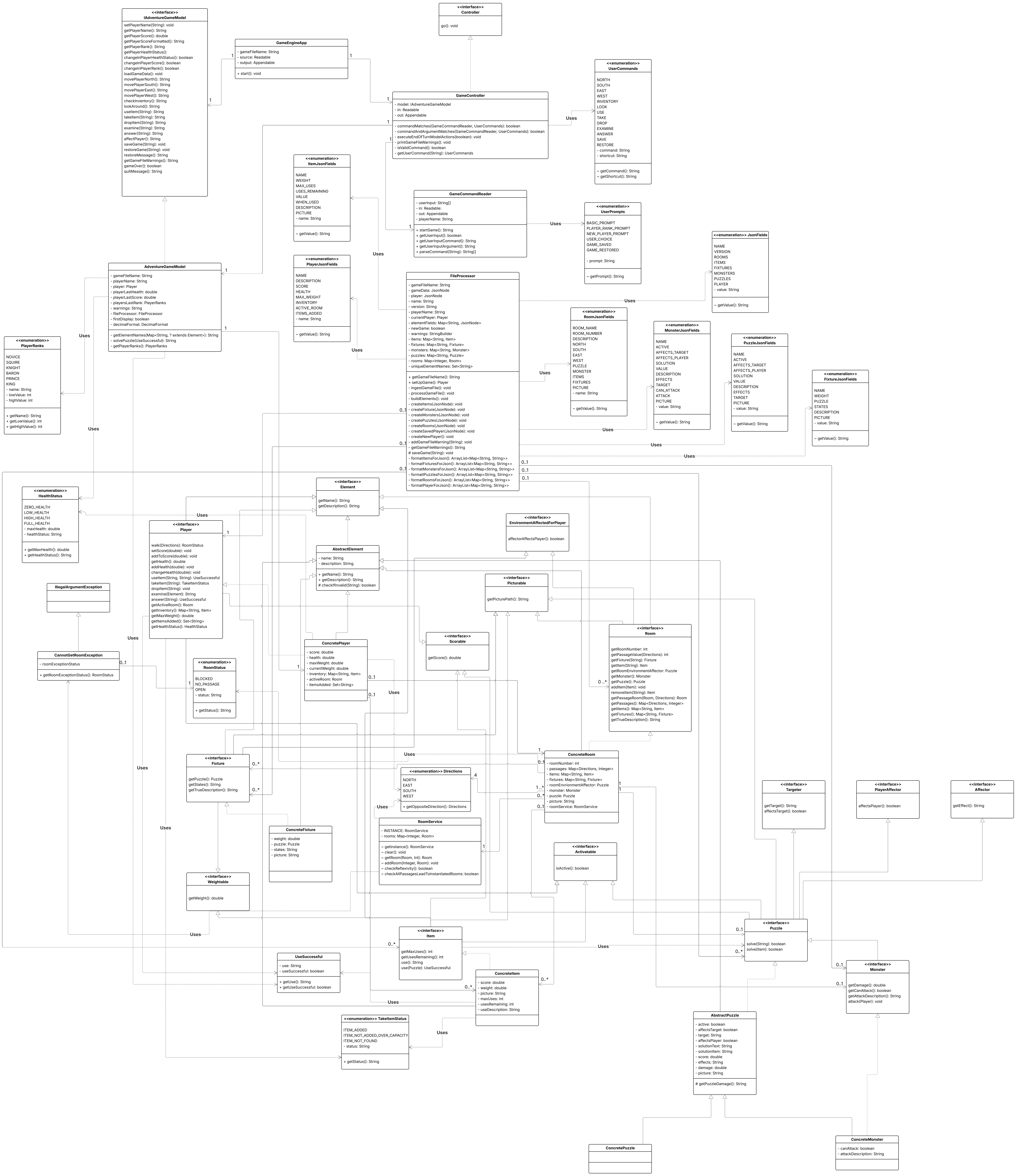
Scenario 2

A player is instantiated in a room with the name “Living Room.” The player is named “Tony” and starts with half health and a carrying capacity of thirteen with nothing in his inventory, so the current weight of the player’s inventory is zero. The player is told that the south passage is blocked by a monster while the west passage is open for navigation. The north and east passages are impassable. There is a TV fixture in the room.

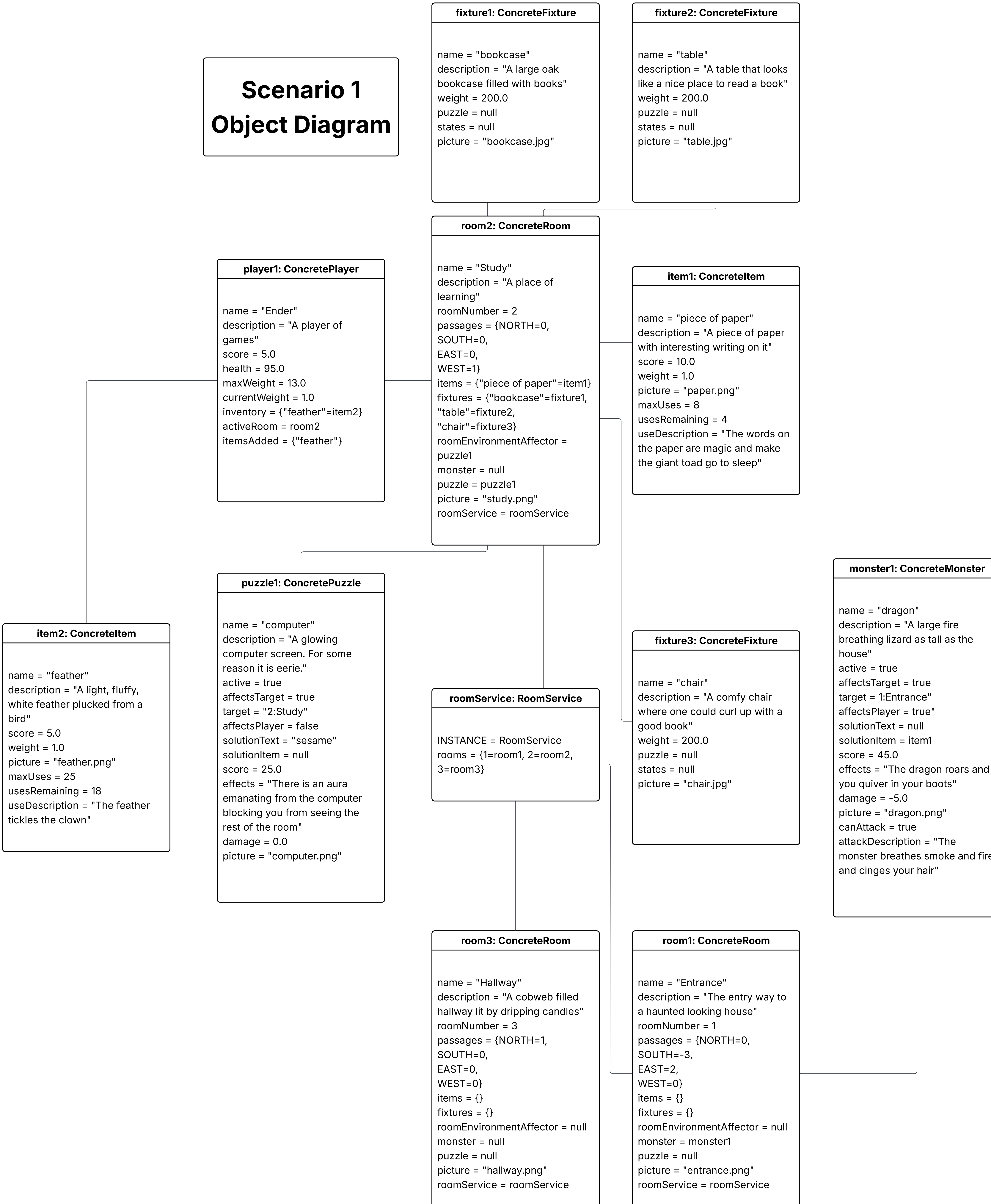
The player navigates through the west passage and enters a new room with the name “Kitchen.” The player inspects the current room and is told that there is one fixture, a refrigerator; an item, a frozen chicken; and a puzzle. The player inspects the refrigerator and is told that it is large and cold. The player inspects the chicken and is told that it weighs two units. The player adds the chicken to his inventory.

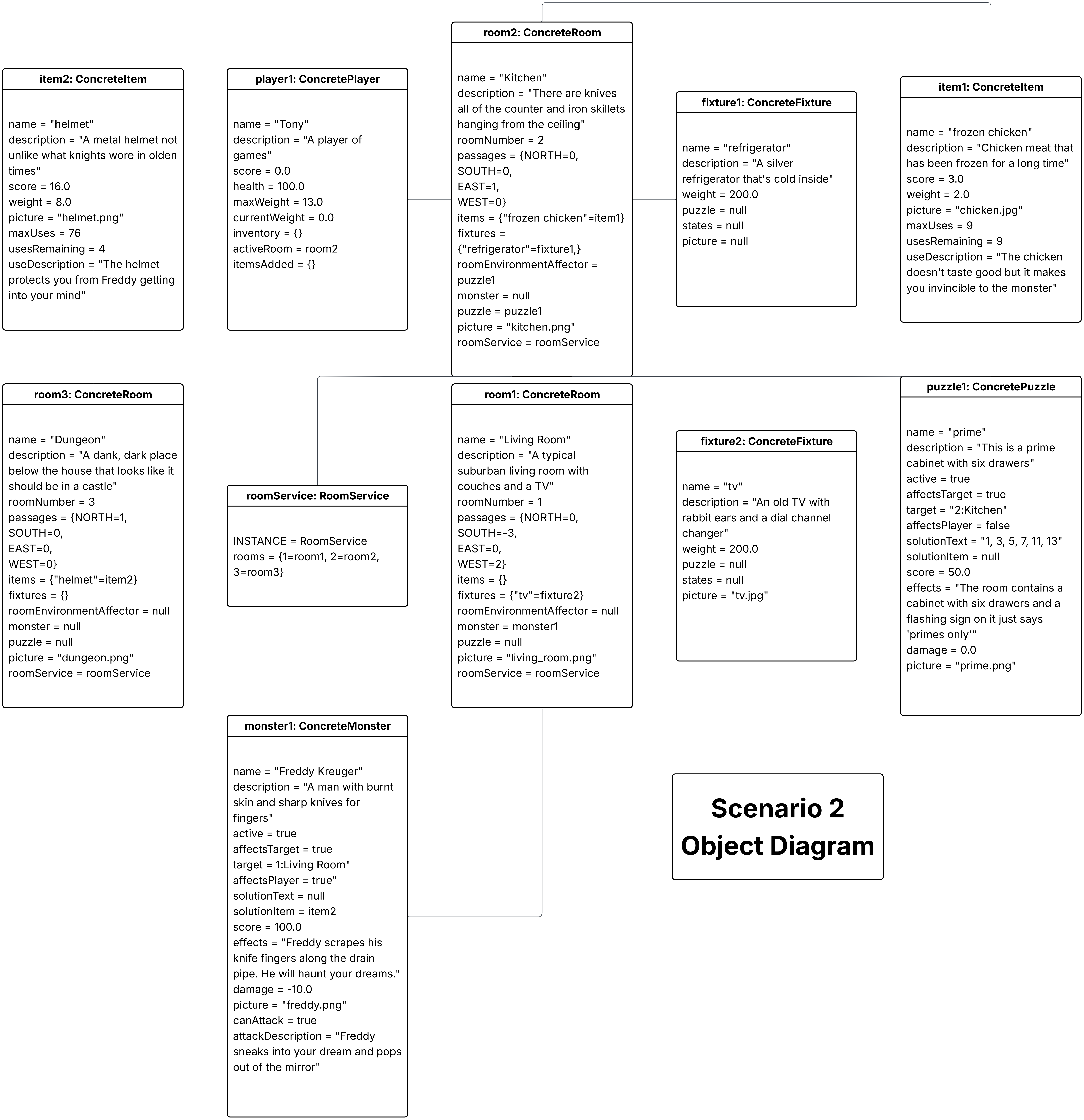
Next, the player inspects the puzzle. The puzzle says, “this is a prime cabinet with six drawers.” The player enters the following sequence as a text answer: 1, 3, 5, 7, 11, 13. The player receives a message that the puzzle is solved and that his points have increased by 50. Finally, the player goes back through the east passage to the Living Room.

Class Diagram

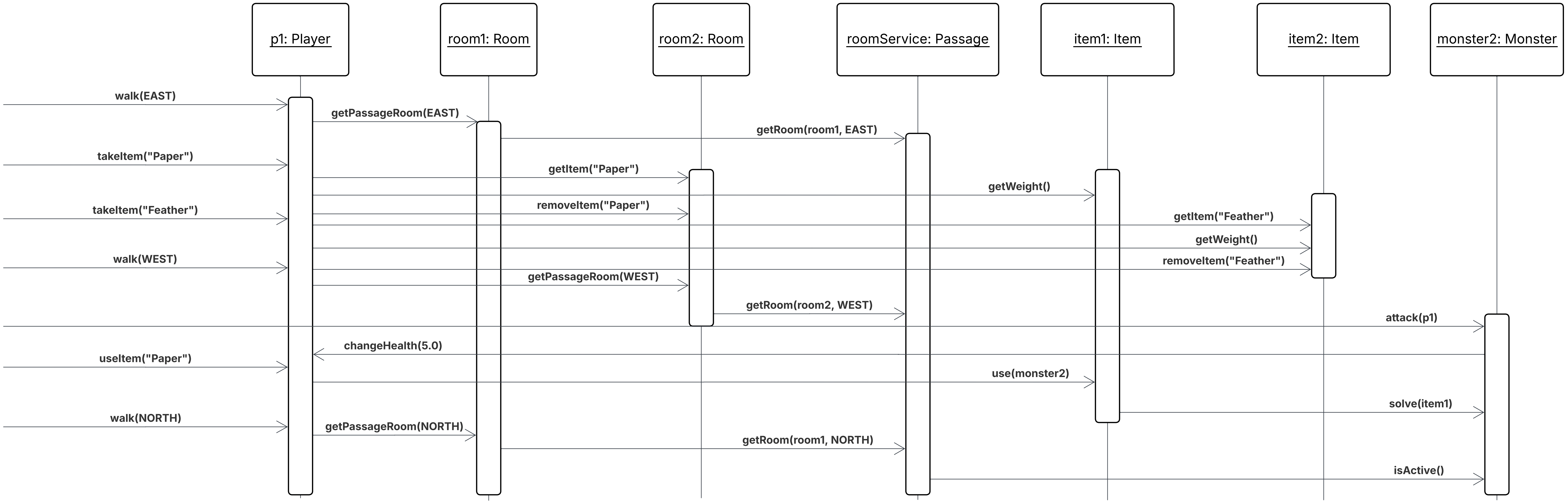


Scenario 1 Object Diagram

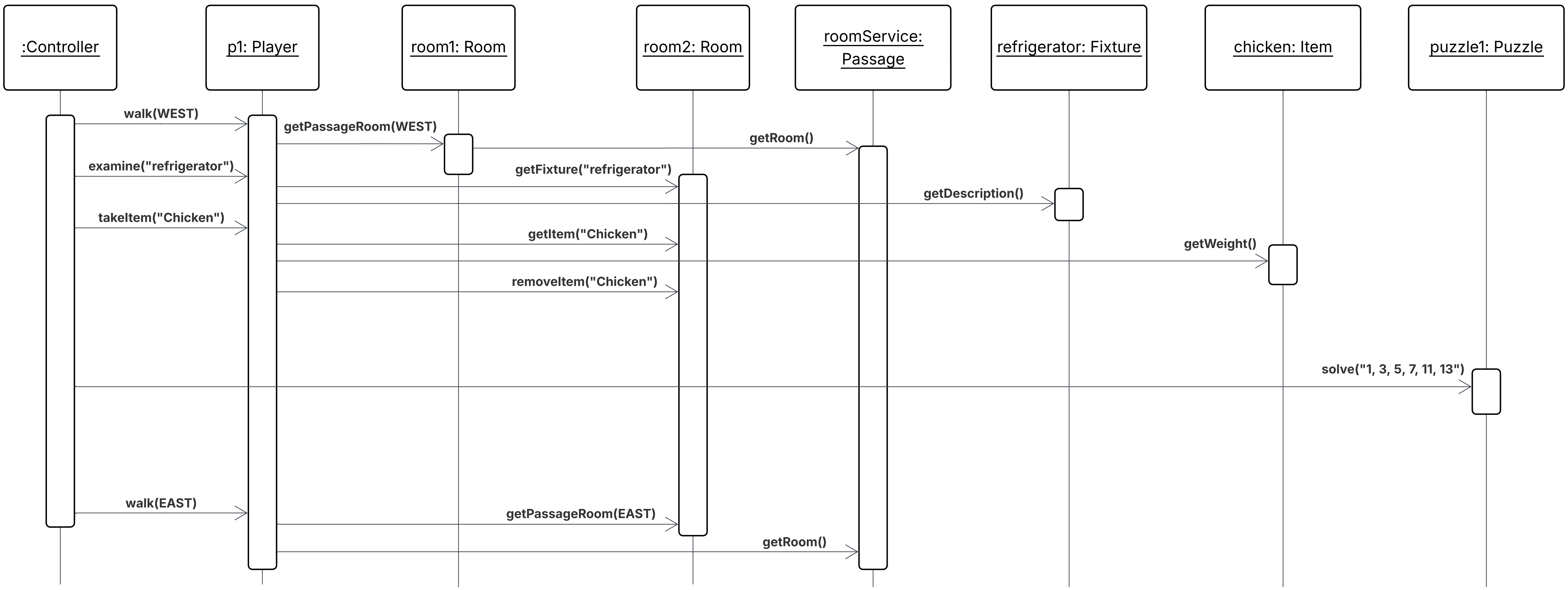



room2: ConcreteRoom
name = "Kitchen"
description = "There are knives all of the counter and iron skillets hanging from the ceiling"
roomNumber = 2
passages = {NORTH=0, SOUTH=0, EAST=1, WEST=0}
items = {"frozen chicken"=item1}
fixtures = {"refrigerator"=fixture1,}
roomEnvironmentAffector = puzzle1
monster = null
puzzle = puzzle1
picture = "kitchen.png"
roomService = roomService
item2: ConcreteItem
name = "helmet"
description = "A metal helmet not unlike what knights wore in olden times"
score = 16.0
weight = 8.0
picture = "helmet.png"
maxUses = 76
usesRemaining = 4
useDescription = "The helmet protects you from Freddy getting into your mind"
player1: ConcretePlayer
name = "Tony"
description = "A player of games"
score = 0.0
health = 100.0
maxWeight = 13.0
currentWeight = 0.0
inventory = {}
activeRoom = room2
itemsAdded = {}
fixture1: ConcreteFixture
name = "refrigerator"
description = "A silver refrigerator that's cold inside"
weight = 200.0
puzzle = null
states = null
picture = null
item1: ConcreteItem
name = "frozen chicken"
description = "Chicken meat that has been frozen for a long time"
score = 3.0
weight = 2.0
picture = "chicken.jpg"
maxUses = 9
usesRemaining = 9
useDescription = "The chicken doesn't taste good but it makes you invincible to the monster"
room3: ConcreteRoom
name = "Dungeon"
description = "A dank, dark place below the house that looks like it should be in a castle"
roomNumber = 3
passages = {NORTH=1, SOUTH=0, EAST=0, WEST=0}
items = {"helmet"=item2}
fixtures = {}
roomEnvironmentAffector = null
monster = null
puzzle = null
picture = "dungeon.png"
roomService = roomService
roomService: RoomService
INSTANCE = RoomService
rooms = {1=room1, 2=room2, 3=room3}
room1: ConcreteRoom
name = "Living Room"
description = "A typical suburban living room with couches and a TV"
roomNumber = 1
passages = {NORTH=0, SOUTH=-3, EAST=0, WEST=2}
items = {}
fixtures = {"tv"=fixture2}
roomEnvironmentAffector = null
monster = monster1
puzzle = null
picture = "living_room.png"
roomService = roomService
fixture2: ConcreteFixture
name = "tv"
description = "An old TV with rabbit ears and a dial channel changer"
weight = 200.0
puzzle = null
states = null
picture = "tv.jpg"
puzzle1: ConcretePuzzle
name = "prime"
description = "This is a prime cabinet with six drawers"
active = true
affectsTarget = true
target = "2:Kitchen"
affectsPlayer = false
solutionText = "1, 3, 5, 7, 11, 13"
solutionItem = null
score = 50.0
effects = "The room contains a cabinet with six drawers and a flashing sign on it just says 'primes only'"
damage = 0.0
picture = "prime.png"
monster1: ConcreteMonster
name = "Freddy Kreuger"
description = "A man with burnt skin and sharp knives for fingers"
active = true
affectsTarget = true
target = 1:Living Room"
affectsPlayer = true"
solutionText = null
solutionItem = item2
score = 100.0
effects = "Freddy scrapes his knife fingers along the drain pipe. He will haunt your dreams."
damage = -10.0
picture = "freddy.png"
canAttack = true
attackDescription = "Freddy sneaks into your dream and pops out of the mirror"
Scenario 2
Object Diagram

Scenario 1 Sequence Diagram



Scenario 2 Sequence Diagram



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